

# LumenCore Evidence Control Plane

Public-Safe Solution Brief Draft for the Sovereign Defense Cloud

Commercial Solutions Opening W912HZ26SC005  
U.S. Army Engineer Research and Development Center  
High Performance Computing Modernization Program

Scope: Unified Service Layer, AI-Powered Orchestration evidence, Secure Data Fabric metadata, and Vendor Lock-In Prevention.

**PUBLIC-SAFE DRAFT - REQUIRED PRICE AND PRIVATE SAM-MATCHED  
IDENTITY ARE NOT INCLUDED**

Website: <https://lumen-core.ai>

Public repository: <https://github.com/robertashworth1986-debug/lumen-core-public>

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## Acronyms and Abbreviations

This list is separate from the five-page proposal body, consistent with the June 9, 2026 Frequently Asked Questions.

| Acronym    | Meaning                                                    |
|------------|------------------------------------------------------------|
| API        | Application Programming Interface                          |
| CSO        | Commercial Solutions Opening                               |
| DoD        | Department of Defense                                      |
| DSRC       | DoD Supercomputing Resource Center                         |
| ERDC       | Engineer Research and Development Center                   |
| GFE        | Government Furnished Equipment                             |
| HPC        | High Performance Computing                                 |
| HPCMP      | High Performance Computing Modernization Program           |
| ML         | Machine Learning                                           |
| OpenAPI    | Open standard for describing HTTP application interfaces   |
| ROM        | Rough Order of Magnitude                                   |
| SDC        | Sovereign Defense Cloud                                    |
| SHA-256    | Secure Hash Algorithm with a 256-bit digest                |
| Zero Trust | Security model that continuously verifies access decisions |

All substantive pages that follow use the full term at first use where practical. Tables and diagrams are included in the five-page body count.

## 1. Mission Gap and Proposed Solution

The High Performance Computing Modernization Program must coordinate hybrid resources, workflows, policy, and data without binding mission decisions to one vendor. LumenCore proposes an evidence control plane that records what was requested, which policy and orchestration path was selected, what artifacts were produced, and whether an independent verifier can reproduce the resulting receipt.

LumenCore is not proposed as a replacement for the five DoD Supercomputing Resource Centers, a complete cloud platform, a cross-domain solution, or a security authorization. It is a modular validation and observability component that can be integrated by the Government or a prime platform provider.

### Mission effectiveness

- Give operators one evidence format across government-owned and commercial environments.
- Detect policy, configuration, adapter, and artifact drift before a result is promoted.
- Preserve adverse outcomes and abstentions instead of reporting only successful runs.
- Allow reviewers to verify a receipt offline without access to the originating control plane.

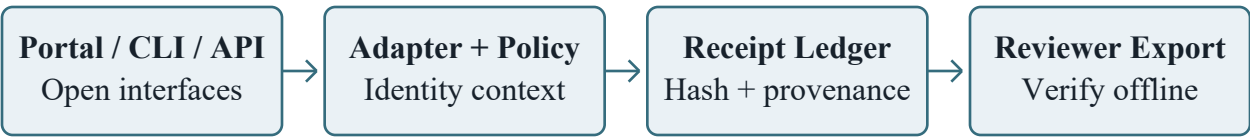
### Focus-area alignment

| Focus area                | LumenCore contribution                                                                                                      |
|---------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| Unified Service Layer     | OpenAPI event and evidence interfaces spanning portal, command-line, and automation clients.                                |
| AI-Powered Orchestration  | Policy-aware evidence hooks record orchestration decisions without replacing the selected scheduler.                        |
| Secure Data Fabric        | Schema, provenance, tag, policy-result, and integrity receipts; workload payloads remain outside the default evidence path. |
| Vendor Lock-In Prevention | Replaceable adapters, portable schemas, offline verification, and no mandatory proprietary cloud service.                   |

### Innovation

The innovation is the application of predeclared acceptance rules, append-only evidence chains, portable verification, and explicit failure retention to hybrid HPC and AI workflow orchestration. The control plane measures decisions across replaceable components rather than requiring a single vendor to own the full evidence path.

## 2. Modular Architecture and Data Boundary



Replaceable adapters: Kubernetes | OpenStack | scheduler | object store | observability

| Component          | Phase II behavior                                                                                                                     |
|--------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| Open adapter layer | Maps selected portal, command-line, scheduler, object-store, and observability events into versioned OpenAPI schemas.                 |
| Policy registry    | Records policy identifiers, versions, inputs, outcomes, and exception reasons; it does not replace Government authorization services. |
| Receipt ledger     | Creates append-only SHA-256 chains for request, decision, artifact, and verification metadata.                                        |
| Offline verifier   | Checks schema, hashes, chain continuity, declared acceptance rules, and missing evidence without network access.                      |
| Reviewer export    | Produces bounded machine-readable and human-readable packets with limitations and failed checks retained.                             |

### Data minimization and sovereignty

The default evidence path stores control metadata, schema references, policy results, hashes, timestamps, and artifact locators. Mission data and model payloads remain in the authorized environment unless the Government defines a narrower approved data flow. Each classification level would use a separately deployed instance and enclave-approved interfaces; this draft does not claim cross-domain transfer or classified-data certification.

### Open replacement boundary

Adapters implement versioned contracts around Kubernetes, OpenStack, workload schedulers, storage, and observability systems. A component may be replaced while the evidence contract and independent verifier remain stable. Government-selected identity and access services provide identity context; Common Access Card authentication is not assumed to be the sole path.

### 3. Phase II Prototype Plan and Feasibility

The following sixteen-week plan is a proposal assumption for Phase II prototype development, not an ERDC-promised schedule. Phase III demonstration and Phase IV implementation costs and activities are excluded from the required Phase II-only price basis.

| Proposed period | Activity                                                                    | Exit evidence                                                   |
|-----------------|-----------------------------------------------------------------------------|-----------------------------------------------------------------|
| Weeks 1-3       | Lock one unclassified use case, interfaces, acceptance rules, and boundary. | Approved interface and test protocol; no production connection. |
| Weeks 4-8       | Build two adapters, policy registry, receipt chain, and offline verifier.   | Schemas, tests, software bill of materials, and build receipt.  |
| Weeks 9-12      | Replay in shadow mode across two approved test environments.                | Retained pass, fail, abstain, missing-data, and drift cases.    |
| Weeks 13-16     | Demonstrate portability, verification, review, and handoff.                 | Government-run verification and limitation register.            |

#### Proposed acceptance protocol

Final thresholds, workloads, exclusions, and test environments would be locked with the Government before scoring. No service level or performance threshold is invented in this draft.

| Acceptance dimension | Proposed measurable check                                                  |
|----------------------|----------------------------------------------------------------------------|
| Integrity            | Every export verifies chain continuity and source-artifact hashes offline. |
| Portability          | One protocol and schema run through two replaceable environment adapters.  |
| Failure visibility   | Failed, abstained, missing-input, and policy-denied runs remain visible.   |
| Reproducibility      | A Government reviewer runs the delivered verifier and documented command.  |

#### Government inputs and assumptions

- One unclassified workflow, approved test data, interface documentation, identity context, endpoints, and security constraints.
- A Government reviewer to approve the protocol and execute the independent verifier.
- No production access or specific Government Furnished Equipment is assumed before ERDC defines the boundary.

## 4. Security, Scalability, Operations, and Risk

### Zero Trust integration boundary

LumenCore consumes identity, device, workload, policy, and environment context supplied by Government-approved services and records the decision evidence. It does not issue credentials, replace access-control authorities, or treat one authenticator as sufficient for every user. Each receipt can bind the applicable policy version, decision result, exception, and verifier outcome.

### Classification and enclave pattern

The architecture can be considered for Unclassified, Secret, Top Secret, and caveated environments through separate enclave deployments and approved interfaces. Phase II should begin unclassified. This is an architectural pattern only; no classified handling, cross-domain transfer, Special Access Program support, or Sensitive Compartmented Information accreditation is claimed.

### Scalability and cost control

- Partition receipts by enclave, mission, tenant, and retention policy without changing the verifier schema.
- Use content hashes and locators instead of duplicating large HPC artifacts in the evidence ledger.
- Expose capture latency, verification time, storage growth, and egress as measured cost drivers.
- Keep adapters and storage replaceable so the Government can compare lifecycle cost and portability.

### Primary risks and controls

| Risk                        | Control                                                                                      |
|-----------------------------|----------------------------------------------------------------------------------------------|
| Undefined legacy interfaces | Use a versioned adapter contract and two bounded interfaces; isolate vendor-specific code.   |
| Sensitive data exposure     | Store metadata and hashes by default; keep payloads in enclave; approve expanded flows.      |
| Performance overhead        | Benchmark capture separately; permit asynchronous finalization where constraints require it. |
| Unsupported scale claim     | Measure Phase II throughput and storage; do not extrapolate beyond observed bounds.          |
| Security accreditation      | Deliver review evidence; do not represent the prototype as authorized to operate.            |

### Operational support boundary

The Phase II scope includes documented escalation, diagnostics, and expert engineering support for the evidence component. A complete Level 3 concierge service for the full Sovereign Defense Cloud is outside this module and would require a qualified integration and operations team.

## 5. Commercial Readiness, Phase II Price Gate, and Evidence

### Commercial approach

LumenCore is modular software plus integration and verification services using commercial technologies and open interfaces. A resultant award is expected to be firm-fixed price. The public repository and website show the development approach; no open-market customer deployment is claimed.

| Evidence                 | What it supports                                                  | What it does not support                                   |
|--------------------------|-------------------------------------------------------------------|------------------------------------------------------------|
| Public repository        | Inspectable builders, tests, schemas, and hash-manifest patterns. | DoD deployment, production readiness, or field validation. |
| Offline verifier pattern | Independent receipt checks and explicit failure reporting.        | Government acceptance or classified accreditation.         |
| Public website           | Authentic project URL and public positioning.                     | Customers, revenue, or realized savings.                   |

### Phase II Rough Order of Magnitude control

**SUBMISSION BLOCKER:** The CSO requires an estimated price for Phase II prototype development only. This public draft includes no price. Labor, infrastructure, subcontractor, travel, indirect cost, profit, payment timing, and firm-fixed-price risk require review before one estimate is inserted privately.

### Final submission gates

| Required finalization             | Current state                                                                             |
|-----------------------------------|-------------------------------------------------------------------------------------------|
| Phase II Rough Order of Magnitude | No price is included. Founder approval is required before private finalization.           |
| SAM identity, address, and status | Insert privately from the active SAM record; verify exact match and contract eligibility. |
| Portal, amendments, and authority | Recheck the live form and amendments; founder reviews all terms and final confirmation.   |

### Bounded next decision

If ERDC considers the module promising, the requested next step is a short technical clarification or solution pitch to lock one representative unclassified workflow, the integration boundary, Government-run acceptance checks, and the appropriate contract or agreement path.

Official project source:

<https://www.erdcerx.org/sovereign-defense-cloud-for-high-performance-computing/>

Funding is not currently available; this market-research lane does not guarantee an award.

Claim boundary: This draft does not claim ERDC selection, available funding, a contract award, Department of Defense deployment, an authorization to operate, classified-data handling, field validation, realized savings, or performance beyond the bounded evidence identified here.